DATA ANALYSIS INTERNSHIP

Day 4 – REPORT

Creating Data Visualization Charts for EDA Insights (Home Assignment)

On Day 4, different **Data Visualization techniques** were used to analyze the **Student Performance Dataset** graphically. Visualization helped in understanding **student study behavior, exam performance, result distribution, and feature relationships** more clearly.

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1. Result Distribution Analysis

Charts were created to analyze the distribution of student results.

Result Categories:

- **Pass**
- **Fail**

Observation:

- The dataset contained slightly more **Fail students (27)** than **Pass students (23)**.
- Students with **higher study hours and better exam scores** were more likely to **pass**.
- Lower performance levels resulted in **Fail outcomes**.

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2. Feature-wise Performance Analysis

Different student performance factors were compared using graphical representation.

Features Analyzed:

- Hours Studied
- Sleep Hours
- Attendance Percentage
- Previous Scores
- Exam Score

Observation:

- **Hours studied** showed a strong impact on exam performance.
- **Exam scores** played the biggest role in deciding **Pass/Fail results**.
- **Previous scores** had a moderate influence on student performance.
- **Sleep hours and attendance** showed weaker influence compared to study hours.

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3. Study Hours Analysis

Visualization techniques were used to analyze the relationship between **study hours and student results**.

Analyzed Areas:

- Hours studied
- Exam score performance
- Pass/Fail outcome

Observation:

- Students with **higher study hours** generally achieved **better exam scores**.
- Students who studied more had a **higher tendency to pass**.
- Regular study habits improved academic performance.

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4. Exam Score Analysis

Charts were used to analyze the distribution of exam scores.

Exam Score Insights:

- Most students scored between **30–40 marks**.
- Very few students had extremely high or low marks.
- Higher exam scores strongly supported **Pass outcomes**.

Observation:

Exam score was identified as one of the strongest factors affecting student results.

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5. Charts Created

The following charts were used for visualization:

Bar Chart

Used to compare:

- Student result distribution (**Pass/Fail**)
- Feature comparisons

Line Chart

Used to analyze:

- Study hour trends
- Exam score variation

Pie Chart

Used to represent:

- Pass/Fail percentage distribution

Histogram

Used to visualize:

- Exam score distribution
- Hours studied distribution

Purpose of Charts:

- Compare student performance features
- Analyze result distribution
- Understand study behavior patterns
- Identify trends and relationships visually

Observation:

- Graphical representation made dataset analysis easier and more understandable.
- Visualization helped identify trends, comparisons, and important performance patterns quickly.

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6. EDA Insights

Important insights identified from the dataset:

- **Higher study hours improved exam performance**.
- **Exam scores strongly influenced Pass/Fail outcomes**.

- **Previous academic scores moderately affected performance**.
- **Attendance and sleep hours showed weaker relationships** with final results.
- Students with balanced study habits generally performed better.

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7. Correlation Visualization Insights

Feature relationships were analyzed through visualization.

Strong Relationships:

- **Hours Studied → Exam Score (0.79)**
- **Exam Score → Result (0.77)**
- **Hours Studied → Result (0.67)**

Observation:

Students who studied more generally achieved **higher exam scores and better outcomes**.

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8. Conclusion

On Day 4, I learned how **Data Visualization** helps in **Exploratory Data Analysis (EDA)**. Different charts were created to analyze **student results, study hours, exam scores, and feature relationships**. Visualization techniques helped in understanding **dataset patterns, comparisons, correlations, and performance insights** effectively.